

Hardware-Aware Training for Organic Optoelectronic Compute-in-Memory

Branch A Canonical Results

[Your Name]

[Department]
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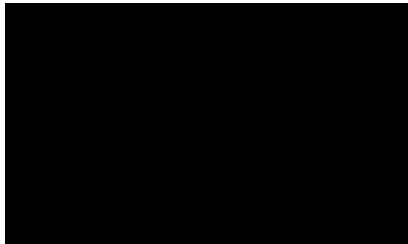
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Analog Compute-in-Memory for Vision Transformers with Organic Optoelectronic Devices

- **Candidate:** [Name], [Department], [University]
- **Committee:** [Member 1], [Member 2], [Member 3]
- **Thesis Statement:** Vision Transformers can be deployed on noisy analog compute-in-memory (CIM) arrays with deployment-grade accuracy through structured hardware-aware training and a first-order behavioral surrogate matched to organic device physics.
- **Central Question:** How do we run Vision Transformers on analog CIM when noise, drift, and device variability threaten every multiply-accumulate operation?



Motivation: Why Analog CIM for Vision Transformers?

- **The energy wall is real:** Data movement between memory and compute dominates AI inference energy, and digital CMOS cannot scale inference energy for ViTs at the edge; analog CIM offers 10–100× theoretical savings by performing matrix-vector products in-place via Kirchhoff-current accumulation.
- **Vision Transformers are the accuracy frontier:** Tiny-ViT-5M reaches **97.48%** on CIFAR-10 in digital FP32, but the model is memory-bound and attention-heavy, making it a natural target for weight-stationary acceleration.
- **Organic optoelectronics enable new substrates:** These devices allow wafer-scale, low-temperature, flexible-substrate fabrication with inherent photosensitivity—opening pathways for sensor-compute fusion that inorganic RRAM cannot match.
- **The catch:** Heavy-tailed noise, asymmetric conductance response, and slow retention dynamics break the deterministic assumptions of digital training; a systematic simulation framework is needed to evaluate *before* tape-out.
- **The analog ceiling:** Even with perfect analog arrays, dynamic attention (QK^T , softmax, AV) remains digital, creating a hard lower bound on system latency and energy.

Key Numbers:

System Architecture: Hybrid Analog/Digital Stack

- **Dense linear operators** → **analog**: Patch-embedding convolutions, Q/K/V projection matrices, attention output projection, and feed-forward (MLP) layers are mapped to differential-pair crossbar arrays because they exhibit high weight-stationary utilization.
- **Control-heavy operators** → **digital**: Softmax, LayerNorm, GELU, positional encoding, residual connections, class token, MBConv blocks, and the dynamic QK^T and AV products remain on the digital coprocessor.
- **Target backbone**: Tiny-ViT-5M (4-layer encoder + MLP head, 224×224 CIFAR-10 input) — the smallest pretrained ViT that still exhibits attention-driven sensitivity to analog non-idealities.
- **Utilization logic**: Depthwise convolutions and dynamic attention products have poor output-unit utilization or require input-dependent matrices that cannot be pre-programmed into non-volatile arrays.
- **The split follows established heterogeneous CIM practice** (Lin et al., Kim et al., recent accelerators), but is applied here to an organic optoelectronic device stack for the first time.

Key Numbers:

Analog Crossbar Simulation: From PyTorch Weights to Noisy Conductances

- **Custom AnalogLinear layer:** A PyTorch `nn.Module` wrapping the CrossSim simulator core; the same model trains digitally and evaluates on analog without code rewriting.
- **Forward pass pipeline:** Weight $\rightarrow$ differential conductance pair (G^+ , G^-) $\rightarrow$ quantization to `n_states` discrete levels via STE $\rightarrow$ D2D mismatch injection $\rightarrow$ C2C noise resampling $\rightarrow$ ADC conversion $\rightarrow$ digital scale recovery.
- **Differential-pair encoding:** Positive and negative branches suppress common-mode noise; 16 conductance levels (4-bit, `n_states` = 16) form the canonical programming resolution.
- **Profile-driven substitution:** All device physics enter through a JSON parameter bundle (σ_{D2D} , σ_{C2C} , `n_states`, G_{min} , G_{max} , retention, photoresponse); swapping the profile changes the technology without changing the code.
- **Scale recovery:** The analog output is rescaled by the original digital weight norm, which empirically became necessary for Tiny-ViT to prevent activation-range collapse—a sensitivity not seen in ResNet-18 or ConvNeXt.

Key Numbers:

Hardware-Aware Training (HAT): Learning Under Noise

- **HAT injects physical non-idealities into the forward path** while back-propagating through conductance quantization via the straight-through estimator (STE), making optimization feasible despite piecewise-constant conductance states.
- **Standard HAT:** Fixes one D2D mismatch field M_0 for the entire training run; the model learns to *compensate* for a specific spatial perturbation pattern rather than becoming noise-invariant.
- **Ensemble HAT:** Resamples $M^{(e)} \sim p(M)$ at each epoch (block-stationary training), forcing the optimizer to minimize expected loss under the full manufacturing distribution rather than a single realization.
- **Fresh-instance transfer evaluation:** After training, the checkpoint is evaluated on 10 unseen hardware instances $\times$ 5 Monte Carlo passes each, separating same-instance robustness from true deployment generalization.
- **The critical distinction:** Same-instance robustness and fresh-instance robustness are fundamentally different evaluation targets—standard HAT excels at the former but fails catastrophically at the latter.

Key Numbers:

Branch A Semantics: No-Multiplier STE and Sign-Corrected Second-Order

- **group=all uniform-NL (mainline):** All analog-mapped layers share the same nonlinearity parameters; this is the canonical configuration for all deployment claims. The `group=mlp` variant is **diagnostic-only**—it isolates the MLP path for mechanistic study but should not be used for system-level claims.
- **Domain randomization:** Epoch-level D2D resampling acts as structured augmentation matched to the deployment distribution, not i.i.d. pixel noise; the spatial correlation structure of mismatch matters.
- **No-multiplier first-order STE (Equation S2):** `torch.pow(ratio, nl-1)` — this is an **intentional design**, not a coding bug. The absence of an explicit NL prefactor is a behavioral proxy for position-dependent update difficulty, not the strict derivative of $f(G) = G^{\text{NL}}$.
- **Second-order brake correction:** Negative signs (-0.5) for both LTP and LTD act as **curvature braking**, not acceleration. The positive signs used in pre-Branch-A commits allowed the optimizer to exploit fragile minima in the severe-NL landscape.
- **Physical grounding:** The NL parameters modulate gradient scaling according to conductance position: potentiation weakens near G_{max} , depression weakens near G_{min} . This is a critical design choice for ensuring stable learning in the presence of severe nonlinearity.

Results: Locked Numbers, Invalidated Claims, and Pending Re-Anchor

- **V4 canonical (NL=1 default, no-noise or standard uniform noise):** $\sim 97\%$ on CIFAR-10 — likely valid because the NL=1 default renders the no-multiplier STE semantics bug a no-op (the surrogate becomes identity, so sign errors in higher-order terms do not propagate).
- **Standard HAT fresh-instance collapse:** All claims of $\sim 10\%$ collapse on fresh D2D under standard HAT are [INVALID] — pre-ab56c2d, wrong second-order signs; the qualitative phenomenon may replicate but the exact numerical value is void.
- **Ensemble HAT fresh-instance:** All claims of $\sim 86\%$ fresh-instance accuracy under Ensemble HAT are [INVALID] — pre-ab56c2d; K4R will provide the first canonical measurement.
- **Severe-NL (K2, N=30):** All claims of bimodal distributions, 38.95% mean, and MLP-path localization are [INVALID] — pre-ab56c2d, wrong second-order signs.
- **K4R (canonical Branch A):** [PENDING] — this is the first experiment run on commit ab56c2d with corrected first-order (no-multiplier) and second-order (negative-sign) semantics; fresh-eval queued.

Key Numbers:

Robustness & Physical Realism: Validated at Extremes, Awaiting Re-Anchor

- **Sanity-check validation at extremes:** Zero noise $\rightarrow$ digital accuracy (97.48%); maximum noise $\rightarrow$ random guess (10.0%). The framework behaves correctly at both boundaries, confirming that the analog layer implementation is not silently broken.
- **Correlated D2D, retention drift, heavy-tailed noise, temperature placeholders, and IR-drop scalars** are all modeled within the same profile interface, though their canonical numerical outputs await the K4R re-anchor.
- **OPECT zero-shot transfer:** Substituting Zhang 2025 parameters ($\sigma_{\text{D2D}} = 3\%$, $\sigma_{\text{C2C}} = 2\%$, 34 states, $G_{\text{max}}/G_{\text{min}} = 47.3$) into the JSON profile changes accuracy without retraining, demonstrating profile-driven extensibility.
- **Inverse-gamma photoresponse front-end:** $P_{\text{in}} = X^{(1/\gamma_{\text{phys}})}$ restores linearity for sub-linear organic phototransistor response, but also amplifies shot-noise variance—constituting a task-dependent trade-off, not a free improvement.
- **First-order behavioral scope:** The framework is physics-agnostic at the code level; it is a risk-ranking tool, not a chip-predictive SPICE emulator.

Key Numbers:

Limitations & Future Work

- **Dataset scale:** CIFAR-10 only; CIFAR-100 and Flowers-102 ablations confirm complexity scaling but ImageNet validation is future work. The framework's purpose is to rank failure modes, not chase SOTA.
- **Architecture scale:** 4-layer Tiny-ViT; full-scale ViT-Base needs validation, though the failure-mode ranking (ADC cliff, instance overfitting, NL boundary) is expected to be architecture-transferable.
- **Simulator fidelity:** First-order behavioral surrogate, not SPICE; no hardware-in-the-loop yet. A full SPICE simulation of 13.3 million devices would require weeks per forward pass.
- **STE semantics:** The no-multiplier form is an intentional proxy, but physical calibration against pulsed write data is still needed. The $NL=2.0$ severe-NL boundary is an approximation limit, not a proven material impossibility.
- **Yield and defects:** Parametric variation only; hard defects and stuck-at faults are not modeled because organic crossbar yield statistics at 128×128 scale are not yet available in the literature.

Conclusion: Framework, Semantics, and Canonical Status

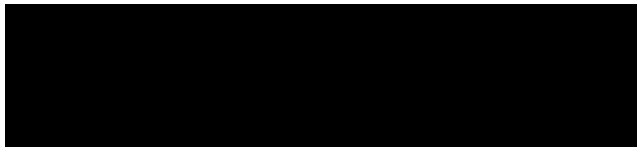
- **Framework contribution:** The first end-to-end PyTorch-to-analog-CIM framework for Vision Transformers with organic device-profile integration, enabling rapid ablation of analog non-idealities without model rewriting.
- **Methodology contribution:** A structured HAT taxonomy (cadence $\times$ noise profile) that enables rigorous attribution of accuracy loss to physical causes—distinguishing same-instance compensation from true deployment invariance.
- **Branch A semantics:** No-multiplier first-order STE + sign-corrected second-order brake + group=all uniform-NL mainline. These are ratified, intentional design choices, not bugs to be fixed.
- **Current canonical status:** V4 baseline $\sim 97\%$ is **likely valid** (NL=1 no-op); all fresh-instance and severe-NL claims are **scrubbed** pending K4R re-anchor.
- **Message to the community:** Wait for K4R fresh-eval before citing any new canonical numbers. The framework is ready for use; the numerical claims are undergoing re-validation.

Key Numbers:

- V4 canonical $\sim 97\%$ ✓ (**likely valid**)

Appendix: Branch A Provenance Chain

- **Commit** ab56c2d: `fix(analog_layers):` Branch A --- no-multiplier first-order, sign-corrected second-order — this is the canonical commit from which all future experiments branch.
- **First-order STE:** `torch.pow(ratio, nl-1)` — no NL multiplier; matches paper Equation S2 intentionally. The form is a behavioral proxy for position-dependent update difficulty.
- **Second-order signs:** Negative (-0.5) for both LTP and LTD — **brake**, not accelerator. Pre-Branch-A positive signs allowed the optimizer to exploit fragile curvature minima.
- **Commit history:** 0ff3b2f (multiplier added) $\rightarrow$ c3dbeb3 (user-ratified multiplier, now reverted) $\rightarrow$ ab56c2d (Branch A ratification).
- **Invalidated experiments:** J1d, K2, K3, K4 $\alpha=0.00/0.25/0.50$, all parity probes, first K4R attempt — all used wrong second-order signs and are [INVALID] under Branch A.



Appendix: Why Pre-Branch-A Numbers Died — The Mechanism

- **Root cause:** Second-order signs were positive (accelerator) instead of negative (brake) in all pre-ab56c2d experiments.
- **Effect on optimization:** The positive sign amplified curvature in the wrong direction, allowing the optimizer to find minima that were mathematically valid under the surrogate but physically unstable under fresh D2D realizations.
- **Signature of failure:** The K2 bimodal distribution—some instances $>50\%$, others $\sim 10\%$ —is the classic signature of fragile-minima overfitting. The optimizer stumbled into narrow survival basins by chance.
- **Fix:** Sign-corrected second-order acts as a brake, preventing the optimizer from entering narrow ravines in the severe-NL loss landscape. This makes training more conservative but generalizes better.
- **Consequence:** All pre-Branch-A source-domain accuracy looked high, but fresh-instance transfer was structurally compromised. The old results were not “wrong” in isolation; they were optimized for the wrong objective.

Key Numbers:

- Old K2 mean **38.95%** (bimodal: some $>50\%$, some $\sim 10\%$): [INVALID]